

Geometrical Problems

Pythagoras' Theorem. A short guide to walk beside you — read as much or as little as helps.
Connection before curriculum. Always.

What this lesson is about

Pythagoras' Theorem is most powerful when the right triangle is not handed to you — you have to spot it inside a larger shape, such as an isosceles triangle or a pair of points on a grid. This lesson helps you find that hidden right triangle, decide what to do with it, and use the answer to solve the original problem. The question you are exploring is: how can a hidden right triangle unlock a geometry problem?

What you are learning to notice

- How to spot a right triangle hidden inside a larger figure.
- How to decide whether the side you want is the hypotenuse (add) or a shorter side (subtract).
- That a straight line dropped from the apex of an isosceles triangle to the middle of the base meets it at a right angle.
- How to take your answer back to the original problem and say what it means.

Moving through it, one step at a time

Sketch the figure and look for a right angle you can use — often by drawing one helpful line. Once you can see a right triangle, decide which length you are after, choose to add or subtract the squares, calculate, and then read the answer back into the original shape. The Scratchpad is useful here for sketching the hidden triangle.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (choosing whether to add or subtract, square roots, and spotting the hidden right angle), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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